

✓ Linear Search

```
%%writefile search1.cpp

#include<iostream>
using namespace std;

bool search(int arr[],int n,int key){
    for (int i=0;i<n;i++){
        if (arr[i]==key){
            return 1;
        }
    }
    return 0;
}

int main(){

    int arr[10]={5,7,-2,10,22,-2,0,5,22,1};
    cout<<"enter the key"<<endl;
    int key;
    cin>>key;
    bool found=search(arr,10,key);

    if(found){
        cout<<"key is present"<<endl;
    }
    else{
        cout<<"key is absent"<<endl;
    }
    return 0;
}
```

 Writing search1.cpp

```
!g++ search1.cpp -o search1
!./search1
```

 enter the key
22
key is present

✓ Binary Search

```
%%writefile search2.cpp

#include<iostream>
using namespace std;

int binarySearch(int arr[], int size, int key){
    int start=0;
    int end=size-1;
    int mid=(start+end)/2;

    while(start<=end){
        if(arr[mid]==key){
            return mid;
        }

        // go to right part
        if(key>arr[mid]){
            start=mid+1;
        }
        else{
            end=mid-1;
        }
        mid=(start+end)/2;
    }
    return -1;
}

int main(){
    int even[6]={2,4,6,8,12,18};
    int odd[5]={3,8,11,14,16};

    int evenindex=binarySearch(even,6,12);
```

```
    cout<<"index of 12 is "<<evenindex<<endl;

    int oddindex=binarySearch(odd,5,8);

    cout<<"index of 8 is "<<oddindex<<endl;

    return 0;
}
```

 Writing search2.cpp

```
!g++ search2.cpp -o search2
!./search2
```

 index of 12 is 4
index of 8 is 1